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Continuous modification in a 3D-printed rocket engine and an additive manufacturing process

Abstract

A rocket motor comprising a non-permeable outer shell, an intra-wall channel between the non-permeable outer shell and an annular section, and a center chamber formed at least in part by a first propellant is disclosed. The first propellant is made up of a multi-grained fuel grain, and the center chamber is further formed by an annular section. The intra-wall channel is configured to permit combustion gases to flux between the non-permeable outer shell and the annular section. The annular section configured to permit combustion gases to flux through pores embedded within the annular section.

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Background/Summary

RELATED APPLICATIONS (1) This application claims priority from the United States provisional application with Ser. No. 63/027,887, which was filed on May 20, 2020, and from the United States provisional application with Ser. No. 63/049,056, which was filed on Jul. 7, 2020. The disclosure of each provisional application is incorporated herein as if set out in full.

TECHNICAL FIELD

(1) The present invention relates to automated fabrication systems, 3D-printable propulsion technologies, solid propellant materials, and methods of producing propellant elements. Specifically, the present invention relates to a method of production of a fuel grain for a solid rocket engine and/or hybrid rocket engine having a fuel grain chamber and geometric shape that induces the flow of oxidizer as it passes through the fuel grain chamber. This invention is not solely limited by fuel grains or hybrid designs, having equally valid use for composite propellant formulations and ensuing solid rocket motors (SRM) with energetic propellant fill comprised of both fuels and oxidizers in an appropriately prepared slurry, subsequently cured or hardened. In both cases, the fuel grain or composite propellant SRM is manufactured via an additive or 3D-printed technique from continuous or semi-continuous feed and mix of ingredients by expedient methods, then expelled into predetermined configurations suitable for rocket propulsion.

BACKGROUND

(2) The 3D-printed rocket engines and gas generators described herein generally encompass propulsion technologies utilizing a pressurized oxidizer stored in a liquid or gaseous state, and a solid fuel. The manufacture of SRMs with blended fuel and oxidizer ingredients are not significantly different in means or methods from hybrid systems having oxidizer(s) separate from printed fuel structures. Prior art systems include both hybrid rocket engines (HRE), liquid rocket engines (LRE), and solid rocket engines (SRE). Reverse hybrid engines (RHE), where the oxidizer is stored in a solid state have also been described, but are much less common. The below background will focus on HREs and composite propellant SRMs due to their expanding relevance in addition to their many distinct advantages over purely liquid or solid systems, including overall simplicity, safety, improved operability, performance, and cost. In addition, the concepts described covering prior art HRE systems are broadly applicable, often encompassing SREs, LREs, RHEs, and other related systems, including the aforementioned composite propellant solid rocket motors. Notably, the term “engine” and “motor” are used interchangeably herein.

(3) The advent of additive manufacturing technology has enabled new methods of designing and manufacturing rocket fuels with much more complex geometries. Using additive manufacturing, for example, solid composite propellant or fuel-rich structures can be designed and printed to include complex chamber geometries that were previously too difficult or impossible to achieve with traditional casting methods. Recent publications demonstrate the use of complex chambers geometries resulting in significant performance increases.

(4) There are several types of additive manufacturing processes that can be used to manufacture fuels. A few common processes include Fused Deposition Modeling (FDM), Selective Laser Sintering (SLS), Stereolithography (SLA, and Laser Spray Modeling (LSM). Relatedly, a laser spray nozzle assembly is described in U.S. Pat. No. 4,724,299. This assembly includes a nozzle body, a housing, a cladding powder passage, and a cladding powder supply system that is operably associated with the passage such that the powder exits the opening coaxial with a laser beam. Notably, while FDM, SLS, SLA, and LSM processes represent some of the most common and most inexpensive forms of additive manufacturing, in many cases these processes provide for inferior material densities, limited scaling capabilities, poor repair capabilities, mechanical joint limitations, and/or machinability issues.

(5) Prior art methods and processes are further known to produce excessive fabrication waste, require excessive labor, expend excessive energy and, in general, suffer from diminished production efficiencies. Prior art fabrication and manufacturing methods also typically alter the coefficient of thermal expansion, thereby introducing heterogeneity into thermal expansion rates, causing stress fractures, cracks and material fatigue such as bending or bowing. Further, there is a need for optimizing contact surface areas in order to maximize thermal energy transfer, reduce material requirements and thereby also reduce weight requirements. Finally, there is a need for 3D fabrication with capabilities of the highest thermal effectiveness, highest temperature allowances, highest pressure allowances, highest compactness, highest erosion resistance, and highest corrosion resistance. In SRMs requiring optimized performance for specific rocket propulsion missions, energetic propellant slurry ingredients, whether fuels, oxidizers, binding resins, curatives, or ballistic modifiers, generally are not known to be adaptable to, or compatible with, conventional 3D printing or additive manufacturing resins, thermoplastics, or thermosets. Instead, an unanswered need exists to 3D print or additively manufacture a tailored ingredients set that is emplaced into SRM chambers and cases that meet stringent requirements of ballistics, mechanical integrity, storage life, having acceptable hazards for transport and use in multi-varied propulsion duties.

(6) Prior Art Fuel Chamber Designs

(7) Recent studies have investigated a variety of performance enhancing geometries and have demonstrated significant increases in regression rates compared to simple straight chamber geometries. Each of said geometries utilize a different technique to manipulate the oxidizer flow. A

first prior art example, conducted in 2014 at Pennsylvania State University (PSU), concerns a composite fuel system utilizing a printed acrylonitrile butadiene styrene (ABS) tabulator structure cast with paraffin (See, Arnold et. al., *Formulation and characterization of paraffin-based solid fuels containing swirl inducing grain geometry and/or energetic additives*, 2014). Regression rates for the turbulator were measured using conventional methods. In this case, the added turbulator was shown to increase regression rates by an average of 35%, although a small sample size of testing was performed.

(8) In a related publication, star swirl fuel chamber designs using an acrylic fuel and gaseous oxygen were employed, achieving regression rates of an average of 250% greater over the standard cylindrical straight chamber design (Id.). This effect was seen to increase with each turn per inch for each swirl design tested.

(9) Additional star swirl chambers are known in the art that utilize ABS and N₂O. The related publications describe notable increases in regression rates over straight chambers for similar oxidizer mass fluxes. Further, these publications demonstrate a potential regression rate increase of up to 67% for the star swirl over straight chamber designs (See, B. R. McKnight et al., *Advanced hybrid rocket motor propulsion unit for CubeSats*, 2015). However, this study was also shown to slightly decrease the combustion efficiency by as much as 15% for some tests.

(10) Relatedly, in 2012 Utah State University (USU) conducted a study investigating the performance of a helical swirl design (See, A. Bath et. al., *Performance characterization of complex fuel chamber geometries for hybrid rocket fuel grains*, 2012). The motivation was to enable larger fuel surface burning area without increasing the length of the fuel grain. The helical fuels were printed in ABS and the oxidizer was N₂O. The helical swirl design increased regression rates by up to 130% when compared with straight chamber fuels at the same oxidizer flux. For post burn regression analysis, USU first sectioned the fuel, then the cross sections were imaged to recreate a computerized mesh of the burned grain. The average radius was then calculated to obtain the average regression rate for the burn.

(11) Conventional cast manufactured hybrids have made use of performance enhancing additives in their formulations. These additives can increase regression rates through enhanced radiation, enhanced heat transfer, as well as increased density. One prior art system has shown that the regression rate of HTPB with GOX oxidizer can be increased by the inclusion of 5% (by weight) 100 nm sized aluminum particles (See, J. C. Thomas et. al., *Hybrid rocket enhancement by micro- and nano-scale additives in HTPB fuel grain*, 2015).

(12) This effort was led by the Jet Propulsion Laboratory in collaboration with the Aerospace Corporation and Texas A&M University. In particular, Thomas et. al. notes that particle size is a key factor in performance enhancement. Micron sized particles increase regression rates by enhancing the radiation heat flux from the combustion flame to the fuel surface. Nano-scale aluminum particles additionally have lower ignition temperatures and higher specific area, releasing energy closer to the surface.

(13) Relatedly, Hassan et. al. showed that the use of Lithium aluminum hydride (LiAlH₄, LAH) and Magnesium hydride (MgH₂, MGH) nanoparticles significantly increased the regression rate of a paraffin-based fuel used with a gaseous oxygen oxidizer (See, A. Hassan, *Characterization of paraffin-based hybrid rocket fuels loaded with nanoadditives*, 2018). Further, Leoni and Vadala showed an increase in performance using a mixture comprised of pure HTPB with a 10% Al—Cu powder composition (See, P. Leoni and G. Vadala, *Innovative metallized nano-sized additives for hybrid rocket propulsion*, 2012). This publication demonstrated an increase in regression rate of up to 37% compared to the standard HTPB fuel.

(14) Prior Art Fuel Grain Structures

(15) Additive manufacturing of hybrid fuels has also enabled the design of hollow grain structures which can then be filled with another fuel. The printed structure can be designed as a matrix that releases pockets of the supplemental fuel. The secondary, tertiary, and quaternary fuels may have

increased performance properties and their use is enabled by the printed grain structure. In addition, thrust profiles can be improved by custom tailoring fuel formulations such that the fuels burn through various sections of varying energy densities.

(16) Lyne et. al. showed that a printed grain matrix structure may be filled with powdered fuels such as coal, graphite and/or aluminum (See, J. E. Lyne, *The use of a 3-D printed, polymer matrix containing pulverized fuel in a hybrid rocket*, 2018). The completed fuel may be roughly 50% ABS and 50% powdered fuel by volume. Voids may be optimized to achieve maximum powdered fuel mass while ensuring the ABS structure does not prematurely collapse. This publication described meaningful increases in thrust levels in addition to novel 3D-printable components. Relatedly, The Aerospace Corporation described a related technique which utilizes a printed grain structure that releases liquid fuel rather than powdered fuel (See, *Aerospace annual rechamber: Creating an agile space enterprise*, 2018). This technique has been stated to potentially increase regression rates by up to 100%.

(17) Additional techniques for achieving performance increases have also been explored. These include using a swirl injector and mixing additives injected into the cast fuel. Some work has also been done exploring the effect of compounding these techniques along with complex fuel chambers. In some cases, this has led to additional performance increases in regression rates. For example, the turbulator grain design can be combined with a paraffin/aluminum fill mixture, in addition to a swirl oxidizer injector, in order to increase regression rates.

(18) Prior Art Composite Propellant SRM Structures

(19) Additive manufacturing of SRMs using slurries of particulate fuel/oxidizer in a liquid resin curable binder is not readily shown or demonstrated even by those well practiced in the science of energetic propellant production, although the prior art of composite propellant thermoset mix/cast/cure is a well-understood aspect of SRM manufacture. The additive manufacturing of SRMs benefits from continuous or semi-continuous feed of propellant slurries from start to finish in the motor 3D printing operation, bound by the slurry characteristics of viscosity, cure rates of the slurry, yield stress, motor size, and other solid/liquid interactions in the composite. Batch mixing of composite slurries may be performed offering a semi-continuous production of SRMs via the additive manufacturing technique, but continuous mixing of the separate ingredients in composite propellant slurries allows control of ballistics, cure properties, and other desirable attributes for SRMs that can be adjusted by design throughout the motor printing process, either within one motor or across multiple motors made while continuously controlling rate and placement of the propellant, in molds having near net shape inserted into motor cases, or into the motor case itself. This novel flexibility and adaptability for SRM production not only enhances automation and thereby reduces cost, but also enables unique motor performance during its mission that is not producible using conventional cast/cure techniques.

(20) Prior Art 3D Printing and Testing Capabilities

(21) Additive manufacturing, or 3D printing, refers to a variety of different processes. The most common method of additive manufacturing for printing complex fuel chambers in recent studies has been fused deposition modeling (FDM). In the FDM process, the fuel is printed layer by layer. In the FDM process, a roll of polymeric filament is continuously fed through a hot end, which melts the material and bonds the material to the previous layer. Once a layer is completed, the extruder raises to complete the next layer. This process repeats for each layer until the final layer is completed.

(22) Advantages of FDM include low cost, ease of use and the availability of open source hardware and software. Ideal FDM printers are capable of printing very fine structures, for example, ABS fuel filaments. ABS has a high print temperature of 240 degrees Celsius, and requires a controlled thermal environment to prevent warping. In addition, heated beds are necessary to ensure that the ABS will adhere to the build plate, and maintain sufficient temperature to prevent premature cooling and warping.

(23) In one example, the MakerBot 3D printer from Johnson and Johnson 3D Printing Laboratory may be used, but it does not contain heated beds and is limited to printing PLA filament. Therefore, a dedicated 3D printer for producing ABS fuels is often preferable. For example, a Creality Ender 3 with a glass heated bed is known in the art to print the ABS fuels and ABS-type fuels with high efficiency.

(24) Indeed, in recent years additive manufacturing (AM) has been implemented in the printing of rocket components using the above components, and the like. AM has enabled design of fuel grains with complex chamber geometries which are not possible or very difficult to achieve through conventional casting techniques. These geometries can be used to manipulate the oxidizer flow and interactions with the fuel to increase performance of the engine. Indeed, some recent studies using a variety of performance enhancing geometries have shown significant regression rate increases. Prior art examples include but are not limited to hot-fire experiments of printed fuels in addition to additive manufacturing techniques for ABS+Aluminum complex chamber fuels grain, the later resulting in successful tests for a 3D printed ABS+Aluminum hybrid rocket fuel grain.

(25) Although recent developments in additive manufacturing are promising, as described above there are many geometries and configurations that remain unexplored. Performance enhancing mechanisms are largely unknown or under development, and much exploration will be required to perfect fuel grain designs, demonstrate reliability, and explore potential new geometries.

(26) For the foregoing reasons there is a need for additively printed complex rocket and gas generator components. As discussed, there is a need for custom tailored fuel chambers and fuel formulations such that when a solid rocket engine burns through its fuel, it burns through various sections of the fuel, which might be of varying energy densities. For the above reasons, there is also a broad need in the art for small gas generators and small diameter rockets (for example, up to 5 inch-60 inch diameter rockets) that are produced in an increasingly efficient and environmentally friendly manner.

(27) The adaptability of continuously varied and controlled composite propellant formulations that are selectively emplaced as novel and unique grain designs, tailored for burn rate and/or mechanical properties, and automated in a 3D printed mode of production, significantly enhances the state of the art. Especially when the continuous mix of separate fuel/oxidizer/binder ingredients are expressed and emplaced via robotic control, the ensuing SRM becomes a product with heightened quality and enhanced performance, for the thrust vs time characteristics can be controlled, or continuously modified, for each and every motor print operation from start to finish.

SUMMARY

(28) The present invention may be used as a rocket motor, system for generating combustion gases, a system for propelling a munition, and the like. This system may be used to propel numerous objects, including but not limited to projectiles, missiles, bombs, supply containers, small satellites, and research stations. Notably, the present invention provides many advantages, including the ability to construct a device with multiple components in a single integrated manufacturing process that is continuously monitored or adjusted. Furthermore, a wide variety of propellant viscosities and multi-grained fuel grain geometries are contemplated relative to conventional systems. Importantly, the present invention permits a user to extrude both sensors and propellants together in a single continuously monitored, continuously modifiable additive process.

(29) In some embodiments, the present invention includes a rocket motor with a non-permeable outer shell, an annular section, an intra-wall channel between the non-permeable outer shell and the annular section, and a center chamber formed at least in part by a multi-grained fuel-rich grain or composite fuel/oxidizer structure and the annular section. Further, the invention uses an intra-wall channel configured to permit, in some embodiments, combustion gases to flux between the non-permeable outer shell and the annular section. "Combustion gases" may refer to pre-combustion or post-combustion gases such as oxidizer, gaseous catalysis, gaseous fuels, and the like. In other embodiments, the intra-wall channel permits gaseous exchange between various sections of the

rocket. Further, the annular section is configured to permit gases to flux through pores embedded within the annular section.

(30) In some embodiments, the invention also includes an infusion element positioned at a distal end of the center chamber proximate to the payload section. Notably, the center chamber has a substantially circular cross-section. In some embodiments, an infusion element is further adapted to infuse a catalyst and/or multiple oxidizers into the center chamber, the infusion element further adapted to modulate the oxidizer to catalyst ratio during operation of the rocket motor. Further, the infusion element may be remotely manipulated by a user such that thrust and burning rates are incrementally and remotely controllable by a user.

(31) In other embodiments, the annular section is shaped to a specified geometry such that combustion of fuel in the center chamber generates a desired ballistics profile. In some embodiments, combustion gases are drawn into the center chamber through the permeable annular section via a pressure differential between the center chamber and the intra-wall channel. Notably, the diameter of the pores embedded within the permeable annular section are determined based on the desired ballistics profile, the desired ballistics profile being calibrated based on varying amounts of combustion gases or alternate elements selected by a user.

(32) In other embodiments, a tertiary chamber **125** is coupled to the intra-wall channel and may house multiple types of oxidizers, catalysts, incendiary fluids, and the like. Notably, in addition to the herein described “rocket fuels”, “rocket fuel” may also refer to multiple types of oxidizers, catalysts, and non-solid propellants/incendiary fluids, and the like. Further, in the preferred embodiment a non-permeable outer shell and the annular section are generated using additive manufacturing techniques. In other embodiments, an electrically-controlled multi-grained fuel grain is utilized, the multi-grained fuel grain making contact with and operatively coupled to various sensors **128**.

(33) In other embodiments, a multi-grained fuel grain for a rocket motor is disclosed that includes a plurality of layers of multi-grained fuel grain material, each layer comprising a plurality of concentric circular structures of different diameter fused together to form a central opening therein. Further, in some embodiments 3D-printing of said concentric circular structures are monitored and modified in real-time. In other embodiments the multi-grained fuel grain material includes an ignitable substance wherein the multi-grained fuel grain is configured to permit mixing of heterogeneous materials in order to allow a user to program ballistics in a stepwise and/or continuous fashion. Further, in some embodiments the plurality of layers is stacked and joined securely to form a cylindrical multi-grained fuel grain embedded with sensors, the sensors enabling continuous monitoring of fuel/oxidizer ratios, solids content, and in-flight mechanical properties. In addition, mixing of heterogeneous materials may include continuous mixing wherein continuous mixing comprises mixing via a robotic continuous mixer.

(34) In other embodiments, the multi-grained fuel grain may comprise metal-based or non-metal based fuel. Further, in some embodiments the metal-based fuel comprises tungsten, magnesium, copper oxide, copper, titanium, and/or aluminum; nonmetal fuel comprises boron, as examples. In other embodiments, a method of operating a rocket motor is disclosed that varies the total impulse achieved. This motor includes a substantially cylindrical center chamber, a multi-grained fuel grain rigidly disposed in said center chamber, and wherein said multi-grained fuel grain is extruded by at least one nozzle during 3D-printing. Said multi-grained fuel grain further includes a plurality of concentric layers of ignitable substance, said layers consisting essentially of a cured mixture of polymeric binder and an oxidizer thereof, said layers being separated from each other by a liner, said liner consisting essentially of a cured polymeric material incapable of supporting its own combustion.

(35) In some embodiments, the at least one nozzle is adapted to extrude high viscosity fuels which allow form retention of the grain when printed, said at least one nozzle housed within the at least one print head and wherein the at least one nozzle is configured to move in sync with said at least

one print head. In other embodiments, multiple print heads are utilized during 3D-printing wherein the at least one print head prints inert compositions while the at least one other print head prints energetics. In some embodiments, controllers and/or control algorithms are used to control the at least one nozzle in order to achieve desired deposition volumes and multi-grained fuel grain forms. In other embodiments, the control algorithm comprises software for controlling the supply of raw materials to the nozzles, the software taking into account 3D-printing extrusion factors, said 3D-printing extrusion factors comprising varied raw material viscosities or slurry cure rates in addition to mechanical variance in the operation of different nozzles.

(36) It is a first objective of the present invention to provide a means for a user to control the ballistics profile of the rocket motor in an on-demand fashion.

(37) It is another objective of the invention to provide a 3D-printable rocket that permits flux between a non-permeable outer shell and an interior annular section, utilizing pressure differentials to achieve flux between sections.

(38) It is another objective of the invention to improve age-life storage conditions, and to provide a means for continuously monitoring additive processing steps.

(39) It is another objective of the invention to enable embedded sensors to be integrated into said fuel grains to facilitate continuous modification during additive processing, or to provide a health-monitoring aspect of the SRM while transported, stored, or operated

(40) These and other advantages and features of the present invention are described with specificity so as to make the present invention understandable to one of ordinary skill in the art.

Description

DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A depicts a rocket engine fuel grain supporting continuous modification of fuel/oxidizer ratios, solids content, and mechanical properties with a generally cylindrical shape including a center chamber according to an embodiment of the invention.

(2) FIG. 1B depicts a rocket engine supporting continuous modification of fuel/oxidizer ratios, solids content, and mechanical properties with a generally cylindrical shape including a center chamber, nozzle and central axis according to an embodiment of the invention.

(3) FIG. 1C depicts a rocket engine fuel grain section with a generally cylindrical shape including a center chamber, nozzle, and helical deposition pattern according to an embodiment of the invention.

(4) FIG. 2A depicts a rocket engine fuel grain section supporting continuous modification of fuel/oxidizer ratios, solids content, and mechanical properties with a generally cylindrical shape including the shell, center chamber, and annular sections and other elements according to an embodiment of the invention.

(5) FIG. 2B depicts an exploded view of generally cylindrical rocket shape including a shell, a pre-ignition section, a post-ignition section, and a payload section.

DETAILED DESCRIPTION OF THE INVENTION

(6) The present invention is directed towards solid and hybrid-fueled rocket motors, solid fuel grains **100**, and methods of production for a rocket engine. In some embodiments additive manufacturing is utilized to construct a novel solid and/or hybrid rocket propellant grain structure. The present invention also comprises methods for using said grains in the operation of a hybrid and/or solid-fueled rocket engine. Notably, these processes are distinct from generalized spraying, wherein insulation is sprayed directly onto rockets. In additional embodiments, additive manufacturing may be utilized to embed an ignition structure or formulation characteristic in order to enhance flame spreading upon ignition on the propellant surface within the rocket motor.

(7) Regarding fuel grains **100**, in some embodiments a solid fuel grain may be manufactured by

various means including a deposition type additive manufacturing apparatus. In the preferred embodiment, the disclosed method of production includes a building platform capable of moving in multi-axis XYZ directions, controlled by step motors. Said step motors are used to modulate the three-dimensional construction of the rocket engine, the rate of fuel grain deposition, and other factors.

(8) Further to the above, the present system is used for printing or otherwise additively manufacturing a solid rocket motor and related rocket motors incorporating solid rocket motor components. As described above, the device is additively manufactured layer by layer on a platform or bed and, in the preferred embodiment, benefits from continuous modification or monitoring. The device includes a propellant or gas-producing energetic material or propellant material processed via continuous modification, sensors that are operatively coupled to the energetic material, and a shell **104** encapsulating the energetic material, sensors, and the like.

(9) Additional details of fuel grains, including their fabrication, are described and claimed below. Preferably, as shown in FIG. 1A and FIG. 2A, the fuel grain section has a generally cylindrical shape and defines a center chamber **102**. In the preferred embodiment, the center chamber **102** has a substantially circular cross-section, but the center chamber **102** may have other geometries such as a star or polygon, without departing from the scope of the present invention. Notably, in some embodiments the fuel grain section may include multi-grained fuel grains, for example a solid propellant comprised of continuous or discrete regions of fuels and oxidizers in liquid, solid, or even gel states.

(10) In the preferred embodiment, the fuel grain section comprises a multi-grained fuel grain. Said multi-grained fuel grain permits variability and control of ballistics, incrementally and infinitely variable with respect to thrust and burning rates. Multi-grained fuel grains permit a user to “program” the ballistics of a rocket among any chosen aspect of solids blends or burning rate catalysts. It is often more practical, arguably, to go from idle to on, or thrust sustain to thrust boost and back again, with solids blend variations or addition of catalysts. Thus, multi-grained fuels are of central importance.

(11) In the preferred embodiment, the herein contemplated invention comprises nozzles **103**, said nozzles **103** generally contained within a print head(s). Multiple print heads are contemplated, one to tailor the inert composition and the other to be focused on the energetic itself. The nozzles **103** are used for extruding or otherwise depositing material for the components of the device. Raw materials for the components may be stored in reservoirs that are operatively coupled to supply materials to the respective nozzles **103**. Notably, in some embodiments, the nozzles **103** and print heads are adapted to extrude high viscosity fuels or composite slurries, in addition to controlled cure-rate thermosets to allow maximum grain design flexibility and optimized throughput.

(12) In some embodiments, the nozzles **103** may have different configurations, and may require different treatments of their respective materials. Further, the print heads are adaptable for use with various nozzle diameters and/or nozzle flow feed mechanisms (such as different rates of feeding raw material) in order to extrude different materials layer by layer. In other embodiments, the present invention is configured to handle materials of heterogenous viscosity. Further, processing speeds are modifiable on-demand and the present invention is generally adapted for use with a wide range of operable speeds. Further, the nozzles **103** may handle a wide range of material volume, and/or may deposit materials of very high-viscosity relative to conventional systems.

(13) In some embodiments, the nozzles **103** may move together as a single unit with the print head. Further, supply of material to one or more of the nozzles **103** at various locations may be used to deposit the distinct materials precisely where each material is needed. Alternatively, one or more of the nozzles **103** may be moved separately from the others. In another embodiment, one nozzle **103** successively translocates in space to deposit different types of material in rapid succession, in order to successively build up a single heterogenous layer. In summary, the above processes may be characterized by continuous variation over time through one or multiple nozzles **103**.

(14) To aid in controlling the content of water and/or glycerol (or other solvent) in the extruded material, the energetic material nozzle may be heated to a higher temperature than that in the general environment of the center chamber **102**. In some embodiments, the nozzle **103** may be heated to a temperature that is at least that of the boiling point of the solvent (such as glycerol and/or water) that is in the baseline propellant material, but is below a temperature at which the baseline propellant material decomposes. The decomposition may involve a heat-producing chemical reaction. This allows for removal of solvent material while still preserving the capability of the extruded material to function as a propellant (to later undergo the chemical reaction between the fuel and the oxidizer that produces propellant products such as pressurized gases). In addition, in some embodiments chemical reaction of the fuel and the oxidizer during the manufacturing process is undesirable in that it produces heat and combustion products, which can damage the manufacturing system. In some embodiments the nozzle **103** comprises an energetic material nozzle that may be heated by any of a variety of suitable mechanisms, many of which may be integrated into printer heads. For instance, the nozzle **103** may be electrically heated, with the electrical heating controlled by a suitable a feedback mechanism that controls the temperature of the nozzle.

(15) In some embodiments, temperature control of the pumpable propellant mixture allows initiation of cure, such as with the use of blocking/deblocking curative ingredients, and associated viscosities as deblocked or uninhibited curatives initiate chain extension of the polymeric resin binder at cure onset. This is one means of control for grain structures that require little or no cure oven conditioning time upon completion of the 3D print operation.

(16) Notably, in some embodiments the temperature of mixing may be directly proportional to the viscosity of the baseline materials (also referred to herein as “slurry”) and the thermal stability of the components used in the mixing system. The viscosity of the slurry determines the shear stress generated during compounding, which in turn determines how well various agglomerates are broken down and the individual particles are dispersed. Thus, in some embodiments it is advisable to use higher temperatures for slurries with a higher viscosity and lower temperatures for slurries with a low viscosity.

(17) In some embodiments, the present system is capable of printing with at least four to ten different materials. Further, the present system is capable of mixing heterogenous materials (i.e., various slurries and/or propellants) during the printing process. Said heterogenous materials may include unfilled thermoplastics and filled thermoplastics with natural fibers, carbon fibers, minerals, and/or metal particles. In some embodiments, continuous mixing via a robotic continuous mixer and continuous extrusion of a given material is contemplated. For example, in one embodiment a slurry material, or the like, may be continuously extruded into a given shape with a mandrel, effectively hybridizing the herein disclosed novel additive manufacturing technique with a more traditional “mix and pour-type” technique. This is permitted by use of a continuous mixer robot, various other robotics, and in-process continuous monitoring.

(18) As described above, in some embodiments additive manufacturing of solid and/or hybrid rocket motors is accomplished using one or more print heads to deliver both inert and energetic materials. Notably, in some embodiments gas may become trapped in various voids during construction. As a consequence, one may utilize an inert shielding gas in order to enhance combustion. For example, carbon dioxide will contribute oxygen to the combustion reaction whereas argon, an inert noble gas, will not. Further to the above, in some embodiments a heavier-than-air inert or non-nanocomposite aluminum reactive gas may be utilized to cover a print bed and a nozzle of the additive manufacturing apparatus. In yet another embodiment, a 3d printer may be configured such that the print bed is fully enclosed and sealed, and an inert non-heavier-than-air gas, such as nitrogen, infills the print bed chamber.

(19) In some embodiments, continuous monitoring of the additive process is contemplated. Continuous monitoring may concern any step of the additive process, including the steps of post-

cure inspection (i.e., x-ray), clean-up, and/or trimming steps. In some embodiments, the printing process is monitored by a logical controller, and a computer is used as an interface to enter the monitoring conditions (i.e., temperature, screw rotational speed, and material rate of deposition). In still other embodiments, monitoring and modification of the following are possible: fuel/oxidizer ratios, solids/binder ratios, mechanical properties, and/or thermoplastic/thermoset curable formulations. In yet another embodiment, the use of formulations including thermoplastic and passivated nanoscale metallic material permits continuous monitoring and modification during printing.

(20) In one embodiment, continuous monitoring addresses the issue of maintaining the filament's diameter and roundness. Notably, a given filament's diameter and roundness often needs to be monitored with laser micrometers or other optical sensors so that the different processing parameters may be adjusted. In order to produce filaments with constant dimensions and roundness, a nozzle may be coupled with a conveyor belt that pulls away the filament, which is eventually spooled in a winding unit. Depending on the thermal conductivity of the filament, it may be necessary to cool down the filament by water or by air. Most highly-filled filaments have a high enough thermal conductivity that no water cooling is necessary. As a result, continuous monitoring may be utilized to monitor and modify all of such parameters in real-time. Such parameters may also include, for example, the extruder temperatures and rotational speed, the speed of the conveyor belt, and/or the spooling device.

(21) As described above, the disclosed method of production includes a building platform capable of moving in XYZ directions, controlled by step motors in order to modulate various factors, and further may permit a continuous horizontal printing pattern. By printing in a continuous printing pattern, the fuel grain production system can continuously print, thus reducing the total printing time. Notably, printing includes applying successive fuel layers continuously for a predefined distance in the direction primarily parallel to the central axis **115**.

(22) In some embodiments, once the model is created and oriented, a preprocessing system then preferably divides the oriented fuel grain model into two-dimensional layers with defined footprint areas that increase with each subsequent two-dimensional layer until a width of the fuel grain model is achieved. The fuel grain production system receives the fuel grain model that is oriented in a direction of the central axis **115** and already divided into two-dimensional layers. The system then prints the fuel grain by applying at least one fuel layer in a direction primarily parallel to the central axis **115**. Printing may include applying successive fuel layers continuously for a predefined distance in the direction primarily parallel to the central core axis **115**. For example, printing may be accomplished by printing a first layer of the fuel grain in a continuous path filling the determined footprint area of the layer with propellant material and then printing successive layers in a similar manner.

(23) In the preferred embodiment, the herein disclosed rocket utilizes a fuel grain that includes a plurality of layers of fuel grain material, each layer including a plurality of concentric circular structures of different diameter fused together to form a central opening therein. Notably, in the preferred embodiment the fuel grain material includes an ignitable substance. In other embodiments, the plurality of layers are stacked and joined securely to form a cylindrical fuel grain with the central opening of each one of the plurality of layers aligned to form a combustion unit extending axially through the fuel grain and bounded by a combustion surface. In some embodiments, the combustion unit includes a circular shape, an oval shape, an elliptical shape, a cylindrical shape, a polygonal shape, or an irregular shape. Notably, the above-discussed central chamber is formed at least in part by said plurality of layers.

(24) In some embodiments, the fuel grain is configured to permit mixing of heterogenous materials to enhance thrust performance. In some embodiments, heterogenous materials comprise unfilled thermoplastics and/or filled thermoplastics, the unfilled thermoplastics and/or filled thermoplastics being filled with natural fibers, carbon fibers, minerals, and/or metal particles. Further, it is notable

that heterogenous materials may comprise unfilled thermoplastics and/or filled thermoplastics with natural fibers, carbon fibers, minerals, metal particles, and heterogenous propellants. In some embodiments, the fuel grain has tailorable chemistries adaptable to continuous monitoring and/or continuous modification during the additive manufacturing process. In other embodiments, the fuel grain further includes a thermally insulating material surrounding the fuel grain.

(25) Regarding the printing process, the melted materials flow through the respective nozzles **103** per FIG. **1B** and FIG. **1C**. These nozzles **103** then extrude the melted materials along a primary axis **115** in a linear pattern. Continuous printing in such a pattern is not required, but does increase the speed of production. The specific pattern of printing allows for a continuous layer of material to be printed without stopping the print head thereby reducing printing time.

(26) FIG. **2A** and FIG. **2B** show an exemplary rocket engine incorporating the fuel grain **100** as described above. As depicted in FIG. **2B**, the exemplary rocket includes a payload section **122** at one distal end of the rocket body **119**, the body comprising a cylindrical shell **106** housing a series of annular sections **120**. FIG. **2A** depicts said annular sections **120** which, in one embodiment of the invention, may be embedded with sensors **101**. For illustration purposes there are spaces **118** between the annular sections **120**, but in an actual device said spaces **118** may be filled with an electrically operated propellant or energetic gas-generating material. As depicted in FIG. **2B**, the exemplary rocket also includes a pre-ignition section **124** and a post-ignition section **126** at the end of the rocket body opposing the payload section **122**. Further, in some embodiments there is an infusion element positioned at a distal end the center chamber proximate to the payload section **122**, wherein the infusion element is configured to inject an oxidizer into the center chamber.

(27) In some embodiments, continuous modification is directed to modification of fuel/oxidizer ratios, solids content, mechanical properties, and the like. In addition, in some embodiments a retardant layer may act to modify burning of an ignition channel. Further, in some embodiments a retardant layer may partially or completely inhibit burning of an ignition channel wall and/or reduce the rate of burning of an ignition channel wall.

(28) In some embodiments, a first propellant has a slower burn rate than a second or third propellant. In other embodiments, the propellant material may be any of a variety of materials that burn to produce pressurized gases, and that is ignited by using the sensors in the preferred embodiment. One class of propellant materials are materials that produce an electrically activated (or electrically operated) solid propellant (also referred to herein as “electrically-activated propellant” or “EAP”). In some embodiments, rocket combustion components may comprise an oxidizer, a rocket fuel, and a binder. In some embodiments, the oxidizer may be a liquid-based perchlorate oxidizer that includes aluminum perchlorate, barium perchlorate, calcium perchlorate, lithium perchlorate, magnesium perchlorate, perchlorate acid, strontium perchlorate, and/or sodium perchlorate. The rocket fuel may comprise a fuel grain (i.e., a solid propellant and solid fuel) and/or an incendiary fluid. Notably, “liquid propellant” and “incendiary fluid” are used interchangeably herein. Said fuel may be a metal-based or non-metal based fuel. Metal-based fuel may include, for example, tungsten, magnesium, copper oxide, copper, titanium, and/or aluminum. The binder may include casein, methyl cellulose, polyethylene oxide, polyvinyl acetate, and/or polyvinyl alcohol. In other embodiments, the rocket fuel comprises a fuel grain section contained in a rocket body, said fuel grain section having a generally cylindrical shape defining the center chamber, the fuel grain section comprising a solid propellant and a solid fuel about said solid propellant. In some embodiments, the center chamber has a substantially circular cross-section but in other embodiments a star shape or irregular shape is possible. In still other embodiments, the rocket engine may run on liquid oxygen and liquid methane, which is an environmental improvement over the solid aluminum and/or ammonia used in many prior art launch assemblies. Use of the above fuels are not only 3D-printable, they're scalable as well. As is described above, 3D-printing reduces the need for special manufacturing of machine parts, cooling chambers, die casts, and the like.

(29) It should be noted that solid-fueled rocket motors in some cases possess characteristics which

render them preferable to liquid-fueled engines, especially for military applications. For example, solid-fueled rocket engines do not require a complicated systems of valves, pumps, and tubing to move the propellant constituents from their storage tanks to the combustion chamber. Moreover, there is no requirement for the storage, transportation, and handling of great quantities of highly corrosive, toxic, and easily vaporized materials as with many liquid-fueled engines. In some embodiments, after each model is printed, it may be tested using technologies like ultrasound to ensure it is structurally sound.

(30) However, liquid-fueled engines have some important advantages over the solid-fueled engines. In a solid propellant grain, the fuel particles and oxidizer are bound together as a single unit. Once the grain has been ignited, combustion continues until the entire grain is consumed. This makes it difficult to vary the total impulse of the rocket motor. Such devices as head-in reversal ports have not proven completely satisfactory in alleviating this problem. On the other hand, the flow of liquid propellant to the combustion chamber can be readily controlled by valves, variable pump speeds, and the like so that the overall total impulse of the engine can be varied. The flow of propellant can even be completely stopped and then restarted if desired. This characteristic of liquid-fueled engines permits propelling missiles varying distances. Thus, it is readily apparent that it would be very desirable to have a hybrid or solid-fueled rocket motor, such as that disclosed herein, that would permit a wider variation in total impulse, thus combining the advantages of both solid-fueled and liquid-fueled engine.

(31) In some embodiments, a hybrid rocket engine (“engine” also referred to herein as “motor”) permits unique operational advantages. As described herein, in some embodiments an oxidizer is introduced into the solid fuel grain **100** section along a pathway defined by a center chamber **102**. Although, in the preferred embodiment, a fuel grain section may be manufactured in various sizes or dimensions, in an exemplary embodiment, the fuel grain section has an outer diameter of at least 6 inches. Although a fuel grain with any grain diameter can be fabricated, in some embodiments a ratio of about 5:1 (outer diameter to inner diameter) may be used for a hybrid rocket fuel grain.

(32) In the preferred embodiment, the rocket engine further comprise an oxidizer source. As described above, the oxidizer source is for flowing through the combustion port during engine operation. As described above, a shell is further utilized for housing at least the fuel grain and the oxidizer source. Notably, the fuel grain composition is described in detail above, include details pertaining to its geometry, construction and its pyrophoric nature.

(33) In some embodiments, the device may include an electrically-controlled (also referred to herein as “electrically-operated”) propellant and/or a comparable energetic material. Said propellant may be additively manufactured in conjunction with said sensors and/or sensors for producing a reaction in the material and/or for providing continuous monitoring and modification capabilities. The device may also include a shell **106** that is additively manufactured with the other components. The additive manufacturing may be accomplished by extruding or otherwise depositing raw materials for the different components where desired. The sensors may be made of a conductive polymer material, for example using an electrically-conductive fill in a polymer. As described above, FIG. 2A shows annular sections **120** which may be embedded with said sensors and other sensors. The spaces **118** between annular sections may be filled with an electrically-operated propellant in some embodiments.

(34) In some embodiments, the present invention includes a rocket motor with a non-permeable outer shell, a permeable annular section, an intra-wall channel between the non-permeable outer shell and the annular section, and a center chamber formed at least in part by a fuel grain and the permeable annular section. Further, the invention uses an intra-wall channel configured to permit incendiary fluid to flux between the non-permeable outer shell and the permeable annular section. Further, the permeable annular section is configured to permit incendiary fluid to flux through pores embedded within the permeable annular section. Notably, “annular section” and “permeable annular section” are used interchangeably herein. The annular section may be permeable to gas in

the preferred embodiments, and liquids in still other embodiments. In some embodiments, the invention also includes an infusion element positioned at a distal end of the center chamber proximate to the payload section, wherein the infusion element is configured to infuse an oxidizer into the center chamber. Notably, the center chamber has a substantially circular cross-section.

(35) In other embodiments, a refueling chamber is coupled to the intra-wall channel and configured to permit refueling of the incendiary fluid. Further, in some embodiments the incendiary fluid is configured to protectively coat the center chamber through the permeable annular section as the solid fuel grain is burned to create thrust. Further, in some embodiments the non-permeable outer shell and the permeable annular section are generated using additive manufacturing techniques. In other embodiments, an electrically-controlled fuel grain is utilized, the fuel grain in contact with and operatively coupled to the sensors. Notably, the sensors permit continuous monitoring and continuous modification of the combustion gases flux and oxidizer flux such that the user controls the ballistics profile of the rocket motor.

(36) In some embodiments, each structural layer may be formed by deposition of viscous polymers, forming a plurality of solidified adjoining structural spheres of material. In some embodiments, the viscosity of said viscous polymers are optimized and controlled such that operational yield stress is enhanced relative to prior art systems. As for the adjoining sphere structures, the objective is to increase the surface area presented to the flame zone for combustion (in some embodiments, within the center chamber **102**) in a manner that is persistent throughout the hybrid rocket engine operation. In this exemplary embodiment, the surface pattern presented to the flame zone is characterized by a series of projections and depressions. According to other embodiments, the surface pattern may comprise a plurality of ribs, divots, knobs and/or recesses.

(37) As described above, the viscosity of said viscous propellants and polymers are optimized and controlled such that operational yield stress is improved in a given system. Notably, the present invention is compatible with highly viscous materials, and is adaptable to perform across a wide range of viscosities. In the preferred embodiment, the present invention contemplates processing with viscosities that are higher than those described in prior art systems. This results in a substantial advantage relative to standard solid rocket preparation in that high viscosity propellants are now easily adaptable to be cast by the batch method, and other comparable methods. As described above, this advantage is largely related to grain size, but also to the unique processing features described above. In some embodiments, rocket motors are capable of being printed in their final high viscosity form (without substantial variance in viscosity throughout the curing process).

(38) In one exemplary embodiment, each fused stacked layer is formed from a series of fused concentric circular spheres of solidified material featuring a pattern designed to increase the surface area of the combustion surface, as compared to a smooth construction. In one embodiment the center chamber **102** features unique patterns designed to induce oxidizer vortex flow persisting throughout the rocket engine's operation as the fuel is consumed. Notably, various surface features that create a heterogenous and/or elevated surface and increase the surface area of the center chamber **102** are contemplated by the present invention.

(39) In some embodiments, the oxidizer may be an aqueous solution with the oxidizer material dissolved in water and/or glycerol. It has been found that controlling the amount of water and/or glycerol in the extruded material in an on-demand fashion facilitates additive manufacture of the propellant material. Too much solvent in the printed mixture can cause problems with curing the extruded material and having the extruded material properly maintain its shape. Too little solvent can result in problems with the material adhering to other material layers.

(40) The raw propellant material that is used to additively manufacture the propellant material may have any of a variety of other suitable formulations. Broadly, the raw propellant material may include a fuel, a solvent (e.g., water and/or glycerol), an oxidizer, and a binder. The fuel and the oxidizer are the chemically active components that react with one another to produce the pressurized gases. The solvent receives the fuel and other components to allow mixing together of

the components into a flowable material. The binder aids in maintaining the raw material as a unified material that can be cured and/or dried into a solid mass. Further to the above, regarding the oxidizer source, a solid, liquid, or gaseous oxidizer may be utilized. As described above, the oxidizer is for flowing through the combustion port during engine operation. However, in one embodiment of the present system, liquid or gaseous oxidizer may be stored adjacent to the motor section or even remotely on the vehicle. Regardless of the configuration or type of liquid or gaseous oxidizer used, the assembly of oxidizer tank, pressurizing system and associated plumbing is typically referred to as the oxidizer port. In some embodiments, the motor port and the oxidizer port are referred to collectively as a hybrid rocket engine system, sometimes also referred to as a hybrid rocket motor.

(41) In some embodiments, a gaseous oxidizer is introduced into the fuel grain section of the center chamber through one or more infusion elements along a pathway defined by the center chamber causing a thin layer of the center chamber wall to phase change from solid to gas vapor. In some embodiments, using a thermoplastic fuel formulation such as 95% by mass Acrylonitrile Butadiene Styrene (ABS) and 5% nanocomposite aluminum, phase change will occur from solid to gas vapor along the exposed surface area of the solid fuel grain center chamber wall. In some embodiments, the resulting combined fuel vapor and nanocomposite aluminum then mixes with the oxidizer to form a fuel/oxidizer mixture suitable for rocket engine combustion. The resulting combusted reaction mass is expelled at high temperature and pressure through the rocket engine's nozzle at a supersonic speed to generate thrust. In some embodiments, fuel formulations with a ratio in a range from about 75% (ABS)/25% (nanocomposite aluminum) up to about 95% (ABS)/5% (nanocomposite aluminum) are utilized.

(42) In some embodiments, the present invention is independent of the particular propellant composition utilized in fabricating the grains, including the multi-grained fuel grains. Further, as described above a first propellant may comprise a propellant composition including a cured mixture of a polymeric binder, fuel particles, and an oxidizer in some embodiments. As polymeric binders, the polyurethanes, polydienes, polysulfides, polylactams, and the like are satisfactory. In other embodiments, double-base propellant compositions are utilized to satisfactory effect.

(43) In some embodiments, the propellant grain can contain no fuel components other than the binder. However, high energy fuels can, and normally will, be incorporated in the composition. Illustrative of these high energy fuels are powdered metal and metal hydrides as exemplified by aluminum, aluminum hydride, magnesium, lithium, lithium hydride, boron, and boron hydride and alloys of these metals. As an oxidizer, inorganic nitrates, chlorates, perchlorates, and the like such as ammonium perchlorates, ammonium nitrate, alkali and alkaline earth metal chlorates, perchlorates, and nitrates (e.g. sodium nitrate, potassium nitrate, potassium perchlorate) will normally be employed, singly or as eutectics or co-crystals with ingredients having positive oxygen balance

(44) In the preferred embodiment, ammonium perchlorate composite propellant (APCP) is utilized as a first propellant. In other embodiments, an electrically-controlled second propellant is utilized. As is well-known in the art, ammonium perchlorate composite propellant is a composite propellant, meaning that it has both fuel and oxidizer mixed with a rubbery binder, all combined into a homogeneous mixture. In the preferred embodiment, APCP is cast or extruded into shape, as opposed to powder pressing as with black powder. This provides manufacturing regularity and repeatability, which are necessary requirements for use in the aerospace industry. In most embodiments, APCP propellant is composed of ammonium perchlorate, an elastomer binder such as hydroxyl-terminated polybutadiene or polybutadiene acrylic acid acrylonitrile prepolymer, powdered metal, and various burn rate catalysts.

(45) In other embodiments, curing additives induce elastomer binder cross-linking to solidify the propellant before use. In still other embodiments, perchlorate serves as the oxidizer, while the binder and aluminum serve as the fuel. In the preferred embodiment, burn rate catalysts determine

how quickly the mixture burns. The resulting cured propellant is fairly elastic, which also helps limit fracturing during accumulated damage (such as shipping, installing, cutting) and high acceleration applications such as hobby or military rocketry.

(46) In some embodiments, the composition of APCP can vary depending on the application, intended burn characteristics, and constraints such as nozzle thermal limitations or specific impulse. In some embodiments, rough mass proportions tend to be about 70/15/15 AP/HTPB/Al, though fairly high performance “low-smoke” can have compositions of roughly 80/18/2 AP/HTPB/Al. While metal fuel is not required in APCP, most embodiments include at least a few percent as a combustion stabilizer, propellant opacifier (to limit excessive infrared propellant preheating), and increase the temperature of the combustion gases (increasing Isp).

(47) Notably, the possible combinations of solid propellant, liner, and oxidizer are extensive. For example, the layers of propellant can be of the polysulfide-perchlorate type as is described in the art. As is described above, a liner material may also be utilized, the liner comprising be polybutadiene couple with a perchloryl fluoridechlorine trifluoride oxidizer containing about 25 percent to about 50 percent by weight perchloryl fluoride. Polystyrene can also serve as a liner and fluorine as the oxidizer. These are but two specific examples of satisfactory combinations and are by no means limiting. Those skilled in the art are aware of the many organic polymers (practically all) which are hypergolic with the oxidizers mentioned hereinabove and will experience no difficulty in finding numerous satisfactory combinations of polymeric liners and hypergolic oxidizers within the framework and guidance of this disclosure.

(48) In some embodiments, the invention contemplates a rigid or solid cast type form and in which the charge includes an encapsulated propellant sphere of liquid propellant in heterogenous distribution throughout a rigid matrix forming the solid charge. In addition, the invention may include the propellant charge as a combustible material having separate propellant spheres of fluid of different material disposed in a heterogenous distribution throughout a rigid matrix wherein each propellant sphere is separately encapsulated in a shell which is not reactive with the fluid and with the matrix at ambient conditions.

(49) In some embodiments, the geometries of the fuel grain can be varied to achieve a desired thrust profile, in addition to (or alternatively) changing the porosity. In some embodiments, as shown in FIG. 2A, the center chamber **102** geometry directly relates to the intermediate wall structure and the fuel grain. These geometries generate a particular thrust profile that may be dictated by mission parameters or other performance specifications of the engine. Relatedly, in some embodiments, a variety of thrust profiles are generated which correspond to a given fuel grain geometry. It may be desirable to employ different thrust profiles in order to account for different mission parameters, such as burn duration, desired impulse, g-force sensitivity of onboard electronics or life-forms, or timing issues that demand different thrust at different times.

(50) In some embodiments, the infusion element is adapted to modulate the volumetric ratio of combustion gases to oxidizer utilized during combustion. Further, flux of the oxidizer and incendiary fluid is continuously monitored and continuously modified by a user such that the user controls the ballistics profile of the rocket motor. In other embodiments, the permeable annular section is shaped to a specified geometry such that combustion of fuel in the center chamber generates a desired ballistics profile. In some embodiments, incendiary liquid is drawn into the center chamber through the permeable annular section via a pressure differential between the center chamber and the intra-wall channel. In other embodiments, only gases are exchanged through the pores of the annular section. Notably, the diameter of the pores embedded within the annular section are determined based on the desired ballistics profile, the desired ballistics profile being calibrated based on varying amounts of incendiary fluid. In other embodiments, the intra-wall channel is configured to permit combustion gases to flux between the non-permeable outer shell and the annular section and the annular section is configured to permit combustion gases to flux through pores embedded within the annular section.

- (51) In some embodiments, combustion gases are drawn into the center chamber through the annular section by a pressure differential between the center chamber and the intra-wall channel. Notably, in some embodiments the diameter of the pores embedded within the annular section may determine based on the desired ballistics profile, the desired ballistics profile calibrated to varying amounts of combustion gases.
- (52) In some embodiments, a tertiary chamber **125** coupled to the intra-wall channel. In some embodiments, the tertiary chamber **125** is configured to house a protective coating material, the protective coating material adapted to coat rocket fuel elements in the center chamber so as to permit stepwise modulation of ballistics during operation.
- (53) Variation in fuel transmission, created by variations in the size, distribution, and type of constituent passages, allow for the distribution of fuel throughout the center chamber **102**, for example, in accordance with the expected fuel regression rate. Similarly, variation in pore size and fuel transmission enables the creation of fuel grain shapes, similar to those of traditional solid motors. Regression, or the amount of fuel consumed over a given amount of time at a given location along the inside of the center chamber **102**, is related to the initial shape of the center chamber **102**, the initial shape and distribution of fuel throughout the center chamber **102**, the flow of fuel that is injected during the firing of the engine, and the chemical nature of the fuel(s) and oxidizer(s) used.
- (54) In some embodiments, a matrix of different geometries is utilized such that fuel grain motors are modified via a continuous variation capacity, independent of size. Further, as discussed above, the chemical nature of the fuels and oxidizers are of critical importance. The herein described invention contemplates related tailorable chemistries for such fuels and oxidizers. Notably, the present invention contemplates the entire range of the explosive process vs. the inert process. Finally, the printed in situ capabilities described above are such that the final product motor has customizable embedded sensors.
- (55) In some embodiments, the herein described invention may be adapted to wide variations in internal ballistics. In some embodiments, modelling may neglect or approximate processes of secondary importance such as flame propagation through various channel voids. Accordingly, the present disclosure provides different propellant grain configurations for achieving different desired performance characteristics.
- (56) In other embodiments, as described above, additive manufacturing is utilized to embed isolated formulations or structures in the propellant. Further, additive manufacturing is utilized to enhance combustion efficiencies or ballistics in a given system. In other embodiments, additive manufacturing is utilized to create motor designs on demand, without the use of molds or templated tooling elements.
- (57) In some embodiments, when used in military applications, additive manufacturing may be utilized to co-express propellant, explosives compositions, or the like, in order to enhance warhead effectiveness. In some embodiments, a room-temperature cure procedure is utilized wherein energetic formulations are cured at room temperature, thereby resulting in a stress-free motor interface across all anticipated propellant/insulation interfaces. Further, alternate design features are contemplated wherein a variety of room temperatures are utilized. The above uses of additive manufacturing also generally benefit from minimized nucleation and minimized fissure creation during storage and/or operation. As a consequence, a reduced incidence of catastrophic failure is experienced.
- (58) In addition, there are a number of other additive manufacturing methods that can be employed to produce solid and/or hybrid rocket fuel grains according to the present invention and using a formulation of slurry material such as a polymer and/or nanocomposite additive, without departing from the spirit and scope of the present invention, including but not limited to: Stereolithography, Selective Laser Sintering, Powder Bed Printing, and Inkjet Head Printing. For the example, a composition of the fuel grain may comprise about 95% by mass Acrylonitrile Butadiene Styrene

(ABS), a thermoplastic possessing combustion characteristics desirable for hybrid rocket engine fuel, and about 5% nanocomposite aluminum. Fuel having this structure is available from several sources, as known by those skilled in the art. With a Young's Modulus of 2.0-2.6 GPa, ABS is 460 times less elastic than HTPB and 38 times less elastic than paraffin wax, making it an ideal material for fabricating a hybrid rocket fuel grain and the center chamber **102** of the combustion chamber.

(59) In some embodiments of the present invention, a particular method of making a fuel grain for use in a rocket engine is contemplated. In the preferred embodiment, said method permits continuous monitoring, continuous mixing, and continuous modification of the slurry layers by way of multiple nozzles. In some embodiments, the method of making a fuel grain includes compounding a first material suitable as a hybrid rocket propellant, a second energetic material, and a third ignitable material according to a predetermined mixture ratio in order to form a fourth material. Notably, in some embodiments this fourth material may serve as a raw material for use in an additive manufacturing device. In other embodiments, this proprietary method involves operating the additive manufacturing device using the above-mentioned raw material to fabricate a solidified fuel grain comprising a plurality of stacked layers. As described above, the stacked layers may be joined securely and combined into a plurality of stacked layers, the plurality of stacked layers forming a cylindrical fuel grain. Further, the plurality of layers may be aligned to form a combustion unit extending axially through the fuel grain and bounded by a combustion surface. In other words, the plurality of layers may form a central chamber as described above. Notably, the above-described fuel grain may be configured to permit mixing of heterogeneous propellants in order to enhance thrust performance.

(60) Generally, energetic materials suitable for use in the present invention are a class of material with high concentrations of stored chemical energy that may be released. Highly energetic materials include, for example, ultrafine aluminum powder, the particle size of which may vary from micron to nanoscale, including particles that are a composite of aluminum and polymer in nanoscale. As known by those skilled in the art, generally a nanocomposite is a material comprising two or more constituent solids, the size of which measures 100 nanometers (nm) or less. Even though the nano-scale aluminum particle cores are completely encapsulated in a polymer based oligomer coating and thus passivated, there remains the possibility that this highly energetic pyrophoric material can still be reactive with oxygen or water vapor.

(61) In one application, the compounded material is stored within a climate controlled environment. As a further safety measure, in some embodiments after fabrication each fuel grain or fuel grain section may be shrink-wrapped to encase the fuel grain or fuel grain section in a thin plastic film to prevent atmospheric exposure prior to its use in a rocket engine. In another embodiment the fuel grain is spray coated with a polymeric material or paint that serves to prevent atmospheric exposure. According to another embodiment the fuel grain or grain segment is inserted into an air-tight packaging cylinder and a vacuum drawn to remove all air. The packaging cylinder is sealed before it is removed from the print bed chamber.

(62) Exemplary constructions of the solid and/or hybrid rocket fuel grains are engineered to both increase the amount of surface area available for combustion as a means to improve regression rate, to improve specific impulse, to generate an oxidizer vortex flow, and to reduce fuel waste by inducing oxidizer axial flow within the center chamber **102** to allow more time for oxidizer and fuel gases (or oxidizer and entrained liquid droplets) to mix and combust more thoroughly. Any surface area pattern or topology that furthers one or more of these objectives, and is sustainable throughout the fuel grain cross-section (i.e., as one fuel grain layer ablates the next fuel grain layer presents a desirable surface area pattern) is considered within the scope of the present invention.

(63) As described above, as shown in FIG. 1A and FIG. 2A, the fuel grain section has a generally cylindrical shape and defines a center chamber **102**. As depicted in FIG. 2A, the solid fuel grain **100** section is formed via continuous modification forming a stack of layers with each such layer formed as a series of abutting and/or interwoven surfaces of solidified material arrayed around the

center chamber **102**. In one embodiment, a heat gun is used to bond the individual layers. As is known by those skilled in the art, other adhesives may be used to join the grain section.

(64) In some embodiments, the present invention may include an electrically-operated propellant or energetic gas-generating material, additively manufactured together with sensors for producing a reaction in the material. As depicted in FIG. 2A and FIG. 2B, the device may also include a shell **106** that is additively manufactured with the other components. The additive manufacturing process may be accomplished by extruding or otherwise depositing raw materials for the different components where desired. The sensors may be made of a conductive polymer material, for example using an electrically-conductive fill in a polymer. Regarding electrically operated propellants, in the preferred embodiment an electrically operated propellant ignites with the application of electricity (from the sensors) and correspondingly extinguishes with the cessation of electricity, even when exposed to high pressures, though below a high pressure threshold. For example, when exposed to ambient or high pressures within the shell, such as atmospheric pressure, pressures greater than 200 psi, 500 psi, 1000 psi, 1500 psi and up to 2000 psi, the electrically operated propellant is extinguished with the interruption of electricity (e.g., voltage or current) applied across the electrically operated propellant. In other words, without application of electricity, the combustion of the electrically operated propellant is not self-sustaining at high pressures, such as high pressures of 200 psi or more and less than 2000 psi. Thus, the electrically operated propellant is configured for “on” and “off” operation under the described variety of conditions.

(65) In other embodiments, a fuel grain for a rocket motor is disclosed. In the preferred embodiment, the fuel grain includes a plurality of layers of fuel grain material, each layer comprising a plurality of concentric circular structures of different diameter fused together to form a central opening therein. Further, in some embodiments 3D-printing of said concentric circular structures are monitored and modified in real-time. Further, in some embodiments the fuel grain material includes an ignitable substance wherein the fuel grain is configured to permit mixing of heterogenous materials in order to achieve a desired ballistics profile. Further, in some embodiments the plurality of layers is stacked and joined securely to form a cylindrical fuel grain embedded with sensors, the sensors enabling continuous monitoring of fuel/oxidizer ratios, solids content, and in-flight mechanical properties. In addition, mixing of heterogenous materials may include continuous mixing wherein continuous mixing comprises mixing via a robotic continuous mixer.

(66) In other embodiments, the fuel grain may comprise metal-based or non-metal based fuel. Further, in some embodiments the metal-based fuel comprises tungsten, magnesium, copper oxide, copper, titanium, and/or aluminum. In other embodiments, the non-metal based fuel comprises a clean-burning fuel, said clean-burning fuel comprising liquid oxygen and/or liquid methane. In other embodiments, a method of operating a rocket motor is disclosed that varies the total impulse achieved. This motor includes a substantially cylindrical center chamber, a fuel grain rigidly disposed in said center chamber, and wherein said fuel grain is extruded by at least one nozzle during 3D-printing. Said fuel grain further includes a plurality of concentric layers of ignitable substance, said layers consisting essentially of a cured mixture of polymeric binder and an oxidizer thereof, said layers being separated from each other by a liner, said liner consisting essentially of a cured polymeric material incapable of supporting its own combustion. In some embodiments, the liner consists essentially of a cured polymeric material incapable of supporting its own combustion, the liner permitting the user to control the ballistics profile in stepwise and continuous fashion.

(67) In other embodiments, said liner is a member selected from the group consisting of the polydienes, the polysulfides, the polyoxides, polyurethanes, polylactams, polystyrenes, polyethylenes, polypropylenes, and polybutylenes. In some embodiments, the at least one nozzle is adapted to extrude high viscosity fuels, said at least one nozzle housed within the at least one print head and wherein the at least one nozzle is configured to move in sync with said at least one print

head. In other embodiments, multiple print heads are utilized during 3D-printing wherein the at least one print head prints inert compositions while the at least one other print head prints energetics. In some embodiments, controllers and/or control algorithms are used to control the at least one nozzle in order to achieve desired deposition volumes and fuel grain forms. In other embodiments, the control algorithm comprises software for controlling the supply of raw materials to the nozzles, the software taking into account 3D-printing extrusion factors, said 3D-printing extrusion factors comprising varied raw material viscosities in addition to mechanical variance in the operation of different nozzles.

(68) Notably, the sensors may also be made of any of a variety of electrically conductive materials. One suitable material may be a conductive polymer material. This may be a polymer material that includes a conductive material fill, such as electrically-conductive powder or electrically-conductive fibers. The fill material may include carbon and/or graphene powders or fibers. Other possible fill materials may be metal powders or other electrically conductive materials, preferably with a melting point that is greater than an extrusion temperature.

(69) The conductive material may also be an extrudable or solid metal, for example copper wire surrounded by a plastic sheath. The material is passed through heated nozzles **103** (or other dispensers), which melts the plastic sheath, while leaving the metal in solid form. Cut bits of metal may come into electrical contact with one another, to allow the sensors to be produced using this process.

(70) Various numbers and placement of sensors are possible. For example, the sensor arrangement may include sensors in concentric circles, parallel plates, or a grid-like array of plates. Whatever the arrangement of the sensors, there may be electrical leads between the sensors and a power source, such as a battery, that is used for providing power to active the propellant material. The electrical power source may be coupled to a suitable control system for providing electrical power to the sensors when burning of the propellant or energetic material is desired. In some embodiments, the present invention contemplates electrically-controlled fuel grains in contact with and operatively coupled to various sensors, said sensors allowing continuous modification of thrust and burn rate in some embodiments such that the user controls rocket ballistics in a remote fashion.

(71) Further to the above, in some embodiments additive manufacturing is utilized to embed electrical conductors in the propellant so as to reduce ESD hazards. In other embodiments, embedded electrochemical sensors are used to indicate relative stabilizer depletion and/or related health monitoring metrics. Further, in some embodiments embedded pressure sensors are utilized to measure internal off-gassing, internal swelling, pressure buildup, and the like. Monitoring these elements enhances one's ability to monitor age-out indicators and other important indicators of device health. In still other embodiments, additive manufacturing is utilized to embed actuators, such as piezoelectric devices, in various locations around the device. Notably, as described above, one may also modify fuel grain configurations during a mission in order to modulate thrust characteristics during flight. In some embodiments, embedded actuators may be crucial to monitoring the health of these systems.

(72) Relatedly, in some embodiments different controllers and/or control algorithms may be used to control the different nozzles **103**, in order to achieve the desired deposition volume and form in various configurations. For instance, software and/or firmware for controlling the movement of the nozzles **103**, and for controlling the supply of various raw materials to the nozzles **103**, may be configured with software and/or firmware that takes into account the differences in operation of different of the nozzles **103**.

(73) In some embodiments, software and/or firmware may include a plugin framework that may support integrating third party algorithms and software for sensor simulation, vehicle models, autonomy, etc. In some embodiments, the plugins may communicate with each other through a subscription paradigm carried out, for instance, using the Data Distribution Services (DDS) standard. This may enable third party engineers to quickly contribute new functionality through the

plugin model including, e.g., rocket control algorithms.

(74) Further, in some embodiments additive manufacturing is utilized to control software and sensor networks which consequently permit operators to vary the characteristics of the end-use product. Multiple variables may be controlled by the user including: inert materials placement, cure rate, design configuration, ballistics, explosive yield, and other structural properties. Notably, “ballistics profile” is used interchangeably with “thrust profile” herein. As described above, monitoring and control of these variables allows for varied thrust profiles. Catalysts may also be utilized to induce a quick cure. Finally, this process is facilitated by the mixing of elements described above in addition to use of the correct viscosity.

(75) One of ordinary skill in the art will recognize that additional embodiments are also possible without departing from the teachings of the present invention or the scope of the claims which follow. This detailed description, and particularly the specific details of the exemplary embodiments disclosed herein, is given primarily for clarity of understanding, and no unnecessary limitations are to be understood therefrom, for modifications will become obvious to those skilled in the art upon reading this disclosure and may be made without departing from the spirit or scope of the claimed invention.

(76) Although the invention has been shown and described with respect to a certain preferred embodiments, it is obvious that equivalent alterations and modifications will occur to others skilled in the art upon the reading and understanding of this specification and the annexed drawings. In particular regard to the various functions performed by the above described elements (components, assemblies, devices, compositions, etc.), the terms (including a reference to a “means”) used to describe such elements are intended to correspond, unless otherwise indicated, to any element which performs the specified function of the described element (i.e., that is functionally equivalent), even though not structurally equivalent to the disclosed structure which performs the function in the herein illustrated exemplary embodiment or embodiments of the invention. In addition, while a particular feature of the invention may have been described above with respect to only one or more of several illustrated embodiments, such feature may be combined with one or more other features of the other embodiments, as may be desired and advantageous for any particular application.

Claims

1. A fuel grain for a rocket, the fuel grain comprising: a cylindrical shell housing a plurality of layers of multi-grained fuel grain material, each layer comprising a series of annular sections of different diameter fused together to form a central opening therein; wherein the multi-grained fuel grain material comprises an ignitable substance, the plurality of layers stacked and joined securely to form a cylindrical fuel grain with the central opening of each-one of the plurality of layers aligned to form a combustion unit extending axially through the fuel grain and bounded by a combustion surface; the series of annular sections filled with the ignitable substance; a center chamber formed at least in part by the plurality of layers, each layer having a series of abutting and interwoven surfaces of solidified multi-grained fuel grain material arrayed around the central chamber to increase the surface area of the combustion surface; wherein the multi-grained fuel grain includes heterogenous materials comprising ignitable substance, the ignitable substance comprising unfilled thermoplastics and filled thermoplastics to enhance thrust performance; wherein increase in surface area of the combustion surface improves regression rate, specific impulse, generates an oxidizer vortex flow, and reduces fuel waste by inducing oxidizer axial flow within the center chamber to allow more time for oxidizer to mix and combust thoroughly; wherein the multi-grained fuel grain material is in contact with and operatively coupled to sensors of configured for continuous monitoring of oxidizer ratios, flux of oxidizer, flux of incendiary fluid, flux of combustion gasses, solids content, thermoset curable formulations, and in-flight mechanical

properties; wherein the flux of the oxidizer, the flux of the incendiary fluid, and the flux of the combustion gases are configured according to a ballistics profile calibrated based on varying amount of incendiary fluid; wherein the sensors are arranged in concentric circles or parallel plates; wherein there are electrical leads between the sensors; wherein the sensors are utilized to measure internal off-gassing, internal swelling, and pressure buildup; and wherein the sensors allow continuous modification of thrust and burn rate.

2. The fuel grain of claim 1, the fuel grain positioned proximate to a payload section at a distal end of the cylindrical shell.
 3. The fuel grain of claim 1, further comprises an infusion element, the infusion element configured to infuse an oxidizer into the center chamber, the infusion element further adapted to modulate an oxidizer to catalyst ratio.
 4. The fuel grain of claim 1, wherein the series of annular sections is configured to permit combustion gases to flux through pores embedded within the series of annular sections such that combustion of fuel in the center chamber generates a desired ballistics profile.
 5. The fuel grain of claim 4, wherein combustion gases are drawn into the center chamber through the series of annular sections by a pressure differential to achieve flux between the series of annular sections.
 6. The fuel grain of claim 1, wherein said fuel grain is operatively connected to a rocket body and center chamber; wherein the plurality of layers of said fuel grain section forms a cylindrical shape; and wherein the center chamber comprises a substantially circular cross-section or an irregular shape.
 7. The fuel grain of claim 6, wherein the center chamber further comprises a star shape.
 8. A multi-grained fuel grain for a rocket motor, the multi-grained fuel grain comprising; a plurality of layers of multi-grained fuel grain material, each layer comprising a plurality of annular sections of different diameter fused together to form a central opening therein; wherein a 3D-printing of said plurality of annular sections is monitored and modified in real-time; wherein the multi-grained fuel grain material comprises an ignitable substance; wherein the multi-grained fuel grain is configured to permit mixing of heterogenous materials in order to achieve a desired ballistics profile, the plurality of layers is stacked and joined securely to form a cylindrical multi-grained fuel grain embedded with sensors configured for continuous monitoring of flux of oxidizer, flux of incendiary fluid, oxidizer ratios, solids content, and in-flight mechanical properties; wherein the flux of the oxidizer and the flux of the incendiary fluid are configured according to a ballistics profile calibrated based on varying amount of incendiary fluid; wherein mixing of heterogenous materials comprises continuous mixing, and wherein continuous mixing comprises mixing via a robotic continuous mixer; wherein the sensors are arranged in concentric circles or parallel plates; wherein there are electrical leads between the sensors; and wherein the sensors are utilized to measure internal off-gassing, internal swelling, and pressure buildup; and wherein the sensors allow continuous modification of thrust and burn rate.
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